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1.12 (a) The data were obtained observational.

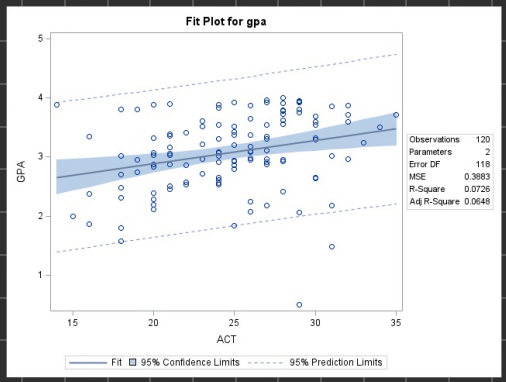
(b) Since the data is not experimental data, making a cause-and-effect conclusion is not valid.

(c) By ChatGPT, dietary habits, stress levels, vaccination status could be some variables affecting both factors.

(d) Construct two groups randomly, one of them exercises a given hour while we do not change the other groups' exercise habit.

1.19 (a) $\hat{\beta}_0 = 2.11405$, $\hat{\beta}_1 = 0.03883 \Rightarrow E(Y) = 2.11405 + 0.03883X$
(Parameter Estimates Table)

(b) Consider R-Square, $R^2 = 0.0726$, which is relatively small, therefore the linear model doesn't fit well.



(c) $\hat{Y} = 2.11405 + 0.03883 \times 30 = 3.27895$

(d) The change in the mean response when the entrance test score increase by one point is $\hat{\beta}_1 = 0.03883$.

2.13 (a) $S_{\hat{Y}}^2 = MSE\left(\frac{1}{n} + \frac{(x_i - \bar{x})^2}{n \cdot \sum(x_i^2) - n\bar{x}^2}\right) = 0.38828\left(\frac{1}{120} + \frac{(28 - 24.725)^2}{120 \cdot 631.158 - 120 \cdot 24.725^2}\right) \doteq 0.0049$

$t_{0.025}(118) \doteq 1.9803$, $t_{0.025}(118) S_{\hat{Y}} \doteq 0.1398$

$\Rightarrow$ mean response 95% C.I. of test score 28

$\Rightarrow [3.2012 \pm 0.1398] = [3.0613, 3.3410]$

(b) $s_{pred}^2 = MSE\left(1+\frac{1}{n}+\frac{(x-\bar{x})^2}{S_{xx}}\right) \doteq 0.3932$, $t_{0.025}(118)\, s_{pred} \doteq 1.2418$

因此, 95% C.I. of test score 28 $\Rightarrow [3.2012 \pm 1.2418] = [1.9594, 4.4430]$

(c) Yes, (b) is wider than (a), since the error of mean is smaller, C.I. is more narrow

2.23 (a)

	DoF	SS	MS	F-value
Regression	1	3.58985	3.58985	9.24
Error	118	45.81761	0.38828	
Total	119	49.40545		

(b) The alternative hypothesis is $\beta_1 \neq 0$ (while $H_0: \beta_1 = 0$),

decision rule : $\{ F > F_{0.01}(1,118) \}$, and yes, we reject H_0.

Codes : github.com/doggydoggy0101/math_project